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2.45 (a) (i) $y'(t) = \frac{d}{dt} f(x(t)) = \frac{d}{dt} \left(\frac{x(t)}{x(t_0)} \cdot f(x(t_0)) \right) = \frac{x'(t)}{x(t_0)} \cdot f(x(t_0)) = f(x(t))$
 $(x(t_0) \neq 0)$

(ii) $y'(t) = \frac{d}{dt} \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau = \frac{d}{dt} \int_{-\infty}^{\infty} h(\tau) x(t-\tau) d\tau = \int_{-\infty}^{\infty} h(\tau) x'(t-\tau) d\tau$
 $= \int_{-\infty}^{\infty} x'(\tau) h(t-\tau) d\tau = x'(t) * h(t) = f(x'(t))$

(iii)

$x(t) \rightarrow [h(t)] \rightarrow y'(t)$

~~or~~

$x(t) \rightarrow [h(t)] \rightarrow [u(t)] \rightarrow y'(t) \Leftrightarrow x(t) \rightarrow [u(t)] \rightarrow [h(t)] \rightarrow y'(t)$

$\Leftrightarrow x'(t) \rightarrow [h(t)] \rightarrow [y(t)] \Leftrightarrow y'(t) = x'(t) * h(t)$

(b)

(i) $x(t) \rightarrow [h(t)] \rightarrow [u(t)] \rightarrow y'(t) \Leftrightarrow x(t) \rightarrow [h'(t)] \rightarrow y'(t)$
 $\Rightarrow y(t) = x(t) * h'(t)$

(ii)

$x(t) \rightarrow [h(t)] \rightarrow y'(t) \Leftrightarrow x(t) \rightarrow [u(t)] \rightarrow [h'(t)] \rightarrow y(t)$
 $\Leftrightarrow \int_{-\infty}^t x(\tau) d\tau \rightarrow [h'(t)] \rightarrow y(t) \Leftrightarrow y(t) = \left(\int_{-\infty}^t x(\tau) d\tau \right) * h'(t)$

$x(t) \rightarrow [h(t)] \rightarrow y(t) \Leftrightarrow x'(t) \rightarrow [u(t)] \rightarrow [h(t)] \rightarrow y(t)$
 $\Leftrightarrow x'(t) \rightarrow \left[\int_{-\infty}^t h(\tau) d\tau \right] \rightarrow y(t) \Leftrightarrow y(t) = x'(t) * \left(\int_{-\infty}^t h(\tau) d\tau \right)$

(c) $y'(t) = \omega_0 \cos \omega_0 t$, $x'(t) = -5e^{-5t} u(t) + e^{-5t} \delta(t)$

$y'(t) = x'(t) * h(t) = -5y(t) + \int_{-\infty}^{+\infty} e^{-5\tau} \delta(\tau) h(t-\tau) d\tau = -5y(t) + h(t)$

$h(t) = y'(t) + 5y(t) = \omega_0 \cos \omega_0 t + 5 \sin \omega_0 t$



(d)

$$x(t) \rightarrow [u_1(t)] \rightarrow [s(t)] \rightarrow y(t)$$

$$h(t) = u_1(t) * s(t) = s'(t)$$

(e) $h(t) = \frac{ds(t)}{dt} = (-3e^{-3t} + 4e^{-2t})u(t) + (e^{-3t} - 2e^{-2t} + 1)\delta(t)$

~~$$y(t) = x(t) * h(t) = \int_{-\infty}^{+\infty} e^{\tau} u(\tau) \cdot (e^{-3(t-\tau)} - 2e^{-2(t-\tau)} + 1) u(t-\tau) d\tau$$~~

~~当 $t < 0$ 时, $y(t) = 0$.~~

~~$$\begin{aligned} \text{当 } t > 0 \text{ 时, } y(t) &= \int_0^t e^{\tau} [e^{-3t+3\tau} - 2e^{-2t+2\tau} + 1] d\tau \\ &= e^{-3t} \cdot \frac{1}{4} (e^{4t} - 1) - 2e^{-2t} \cdot \frac{1}{3} (e^{3t} - 1) + e^t - 1 \end{aligned}$$~~

$$y(t) = x(t) * h(t) = \int_{-\infty}^{+\infty} e^{\tau} u(\tau) (-3e^{-3(t-\tau)} + 4e^{-2(t-\tau)}) u(t-\tau) d\tau + \cancel{e^t u(t) (e^{-3t} - 2e^{-2t} + 1)}$$

~~当 $t < 0$ 时, $y(t) = 0$~~

$$\begin{aligned} \text{当 } t > 0 \text{ 时, } y(t) &= \int_0^t e^{\tau} (-3e^{-3t+3\tau} + 4e^{-2t+2\tau}) d\tau + \cancel{e^t u(t) (e^{-3t} - 2e^{-2t} + 1)} \\ &= -3e^{-3t} \cdot \frac{1}{4} (e^{4t} - 1) + 4e^{-2t} \cdot \frac{1}{3} (e^{3t} - 1) \\ &= -\frac{3}{4} e^t + \frac{3}{4} e^{-3t} + \frac{4}{3} e^t - \frac{4}{3} e^{-2t} = \frac{3}{4} e^{-3t} - \frac{4}{3} e^{-2t} + \frac{7}{12} e^t \end{aligned}$$

2.46 $\frac{dx(t)}{dt} = -3x(t) + 2e^{-3t} \delta(t-1) \Rightarrow e^{-2t} u(t) = \int_{-\infty}^{+\infty} 2e^{-3\tau} \delta(\tau-1) \cdot h(t-\tau) d\tau$

$$= 2e^{-3} \cdot h(t-1)$$

$$h(t) = \frac{1}{2} \cdot e^{3-2(t+1)} \delta(t+1) = \frac{1}{2} e^{1-2t} u(t+1)$$



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$$2.65 \text{ (a)} \quad \phi_{x_1 x_1}[n] = \begin{cases} 0, & |n| > 2 \\ 2, & |n| = 1 \\ 1, & |n| = 2 \\ 3, & n = 0 \end{cases} \quad \phi_{x_2 x_2}[n] = \begin{cases} 0, & |n| > 3 \\ -1, & |n| = 3 \\ 2, & |n| = 2 \\ -3, & |n| = 1 \\ 4, & n = 0 \end{cases}$$

$$\phi_{x_3 x_3}[n] = \begin{cases} 0, & |n| > 2 \\ 1, & |n| = 2 \\ 4, & |n| = 1 \\ 6, & n = 0 \end{cases}, \quad \phi_{x_4 x_4}[n] = \begin{cases} 0, & |n| \neq 5, 0 \\ 2, & n = 0 \\ 1, & |n| = 5 \end{cases}$$

~~$$(b) \quad \phi_{x_1 x_2}[n] = \begin{cases} 0, & n \geq 2 \\ -1, & n = 1 \\ 0, & n = 0 \end{cases}$$~~

$$(b) \quad \phi_{x_1 x_2}[n] = \begin{cases} 0, & n \leq -2 \\ -1, & n = -1 \\ 0, & n = 0 \\ -1, & n = 1 \\ 1, & n = 2 \\ 0, & n = 3 \\ -1, & n = 4 \\ 0, & n \geq 5 \end{cases}$$

$$\phi_{x_1 x_3}[n] = \begin{cases} 0, & n \leq -1 \\ 1, & n = 0 \\ 3, & n = 1 \\ 4, & n = 2 \\ 3, & n = 3 \\ 1, & n = 4 \\ 0, & n \geq 5 \end{cases}$$

$$\phi_{x_1 x_4}[n] = \begin{cases} 1, & -4 \leq n \leq -2 \text{ 或 } 1 \leq n \leq 3 \\ 0, & n \neq -4, -3, -2, 1, 2, 3 \end{cases}$$

$$\phi_{x_2 x_3}[n] = \begin{cases} 0, & n \leq -3 \\ 1, & n = -2 \\ 1, & n = -1 \\ 0, & n = 0 \\ 0, & n = 1 \\ -1, & n = 2 \\ -1, & n = 3 \\ 0, & n \geq 4 \end{cases}$$

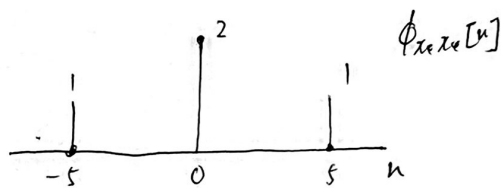
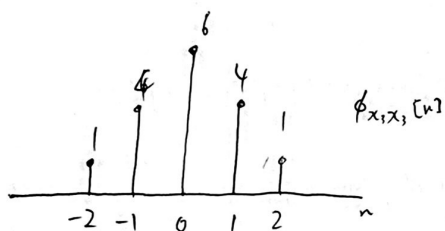
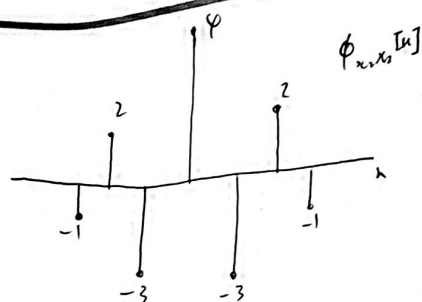
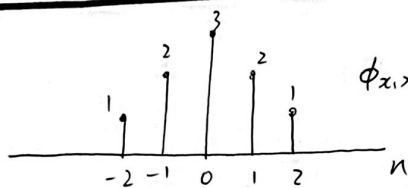
$$\phi_{x_2 x_4}[n] = \begin{cases} -1, & n = -5, -3, 0, 2 \\ 1, & n = -6, -4, -1, 1 \\ 0, & n \notin \{k \mid -6 \leq k \leq 2 \text{ 且 } k \neq -2\} \end{cases}$$

$$\phi_{x_3 x_4}[n] = \begin{cases} 1, & n = -6, -4, -1, 1 \\ 2, & n = -5, 0 \\ 0, & n \notin \{-6, -5, -4, -1, 0, 1\} \end{cases}$$

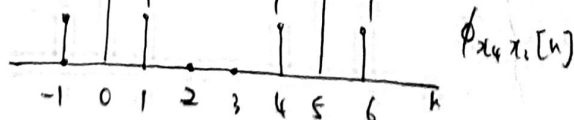
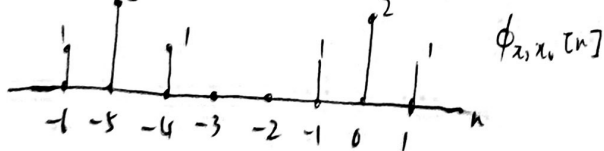
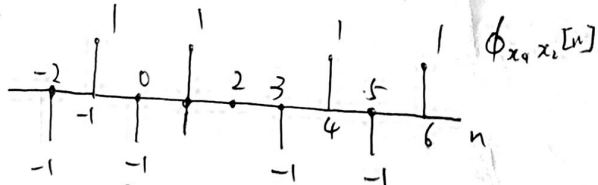
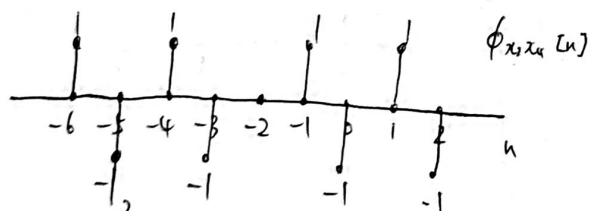
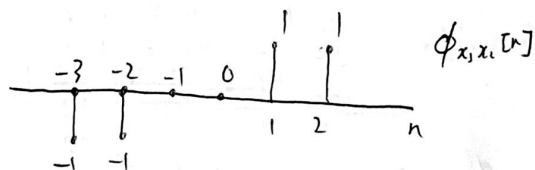
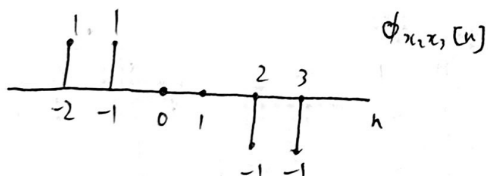
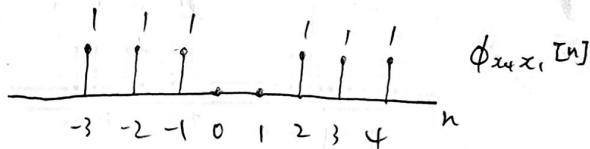
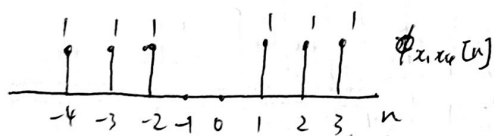
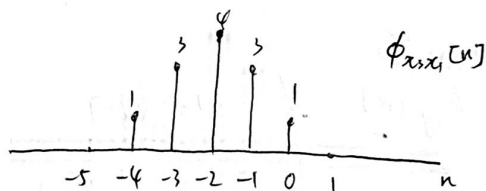
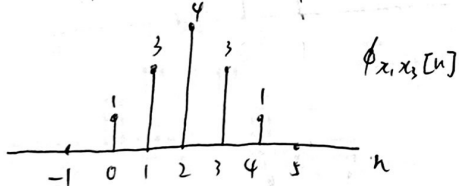
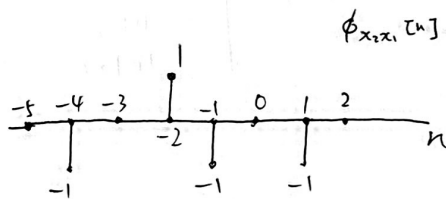
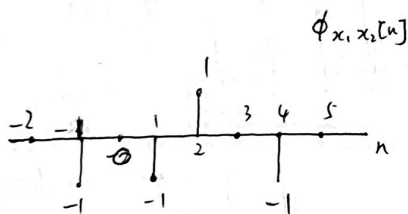


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2.65 (a) (续).



(b)

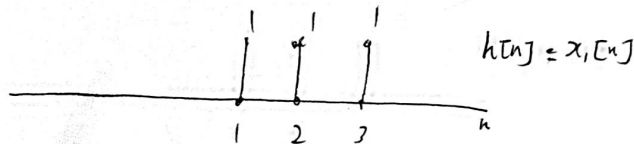
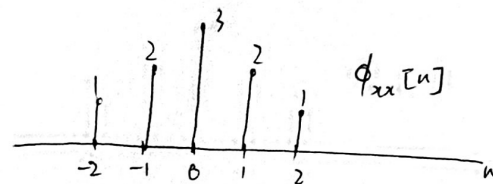


$$\begin{aligned}
 (c) \quad \phi_{xy}[n] &= \sum_{m=-\infty}^{+\infty} x[m+n] y[m] = \sum_{m=-\infty}^{+\infty} \left(x[m+n] \cdot \sum_{k=-\infty}^{+\infty} x[k] h[m-k] \right) \\
 &= \sum_{m=-\infty}^{+\infty} \sum_{k=-\infty}^{+\infty} x[m+n] x[k] h[m-k] \\
 &= \sum_{k=-\infty}^{+\infty} \sum_{m=k}^{+\infty} x[m-k] x[m+n] h[k] = \sum_{k=-\infty}^{+\infty} h[k] \phi_{xx}[n+k] = \sum_{k=-\infty}^{+\infty} h[k] \phi_{xx}[-n-k] \\
 &= \phi_{xx}[-n] * h[n] = \phi_{xx}[n] * h[n]
 \end{aligned}$$

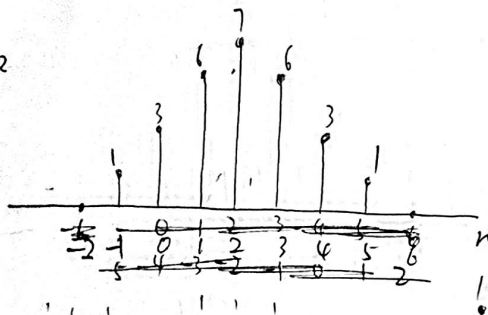
$$\begin{aligned}
 \phi_{yy}[n] &= \sum_{m=-\infty}^{+\infty} y[m+n] y[m] = \sum_{m=-\infty}^{+\infty} \left(\left(\sum_{k=-\infty}^{+\infty} x[m+n-k] h[k] \right) \cdot \left(\sum_{l=-\infty}^{+\infty} x[m-l] h[l] \right) \right) \\
 &= \sum_{m=-\infty}^{+\infty} \sum_{k,l \in \mathbb{Z}} x[m+n-k] h[k] x[m-l] h[l] \\
 &= \sum_{k,l \in \mathbb{Z}} h[k] h[l] \phi_{xx}[n-k+l] \\
 &= \sum_{k=-\infty}^{+\infty} h[k] (\phi_{xx}[-n+k] * h[n]) = \phi_{xx}[n] * (h[n] * h[n])
 \end{aligned}$$

(d)

$$\phi_{xx}[n] = \phi_{x,x}[n] = \begin{cases} 3, & n=0 \\ 2, & |n|=1 \\ 1, & |n|=2 \\ 0, & |n| > 2 \end{cases}$$



$$\phi_{xy}[n] = \phi_{xx}[n] * h[n] = \begin{cases} 10, & n \leq -2 \\ 1, & n = -1 \\ 3, & n = 0 \\ 6, & n = 1 \\ 7, & n = 2 \\ 6, & n = 3 \\ 3, & n = 4 \\ 1, & n = 5 \\ 0, & n \geq 6 \end{cases}$$



$$\phi_{yy}[n] = \phi_{xy}[n] * h[n] =$$

$$\begin{cases} 0, & n \leq -1 \\ 1, & n = 0 \\ 4, & n = 1 \\ 10, & n = 2 \\ 16, & n = 3 \\ 19, & n = 4 \\ 16, & n = 5 \\ 10, & n = 6 \\ 4, & n = 7 \\ 1, & n = 8 \\ 0, & n \geq 9 \end{cases}$$

